

ADS Architecture Review v0.1

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Repo: <https://github.com/alex-place/lantern-os>

Status: current architecture review for Lantern OS after whitepaper, RAG dollhouse, local wallet, dual-boot workforward, and COMET LEAP cash sprint.

DEFINITION

ADS means *Architecture Decision System* for this repo.

It is the operating layer that decides:

- what gets promoted;
- what stays held;
- what becomes a skill;
- what becomes a PDF/report;
- what becomes a cash offer;
- what is unsafe, speculative, or not yet validated.

EXECUTIVE FINDING

Lantern OS is now a coherent local-first control plane, not a skeleton.

The architecture is strong enough for:

- shareholder/operator review;
- RAG-backed memory;
- printable whitepaper distribution;
- manual cash sprint execution;
- dual-boot preparation;
- school/art packet sharing;
- local wallet and invoice tracking.

It is not yet v1.0.0 because the following remain held or operator-action dependent:

- no operator approval for v1 release;
- dual boot install is now prep-scripted, but partition shrink and installer action remain physical operator steps;
- cash sprint is now send-ready, but cleared cash remains \$0 until money actually clears;
- orchestrator dirty state was fixed and pushed in `gm-agent-orchestrator` commit `f4eb6b5`;
- Archive/Wayback/media lane now has an explicit rights gate; full downloads remain held until operator rights/storage review.

ARCHITECTURE MAP

Layer	Current Surface	Path	Review
Control Plane	Lantern OS repo	C:\tmp\lantern-os	PASS: clean pushed master
Memory Plane	RAG Dollhouse	skills/lantern-rag-dollhouse/	PASS: flat file plus hashed assets
Evidence Plane	manifests and evidence records	manifests/	PASS: Bitcoin/Ethereum/governance anchors recorded

Layer	Current Surface	Path	Review
Decision Plane	Bayesian world model	data/world-model/belief- ledger.jsonl	CANDIDATE: ledger exists, needs continuous polling
Money Plane	Local wallet and invoice drafts	data/wallet/	PASS: factual ledger, no fake revenue
Execution Plane	Orchestrator queue	C:\Users\alexp\Documents\gm-agent-orchestrator	PASS: dirty health-check fix pushed in f4eb6b5; cash task queued
Device Plane	Dual boot bundle	dual-boot/	CANDIDATE: prep-ready, install held
Product Plane	Whitepaper, atlas, cash sprint	reports/ and artifacts/	PASS: printable artifacts exist
Learning Plane	Gage school art packet	school-packets/gage-high-intel-art/	PASS: packet exists, privacy-aware framing
Commons Plane	Archive/Wayback/OSS batch	scripts/Invoke-ArchiveCommonsBatch.ps1	CANDIDATE/HOLD: explicit rights gate added; downloads still operator-held

TOP DECISIONS

Decision	Status	Evidence	Reason
Make lantern-os the master repo	promote	pushed master, clean status	one control plane beats scattered repos
Make the RAG dollhouse the memory spine	promote	flat file plus 42 asset manifest rows	gives retrieval a stable body
Make the whitepaper canonical	promote	reports/LANTERN-OS-WHITEPAPER-v0.1.md and PDF	gives the project one printable thesis
Track money in a local wallet first	promote	data/wallet/local-cash-wallet.json	avoids fake revenue and premature payment plumbing
Execute existing offers only	promote	11-day cash sprint and wallet rules	stops offer sprawl
Treat current PC dual boot as prep-ready only	promote/hold	readiness JSON: readyForPrep=true, readyForInstall=false	no unallocated install space yet
Treat iPhone/phones as edge nodes first	hold true dual boot	phone boundary manifests	phone boot paths need device-specific review
Treat offline/server-farm tokens as unmetered capacity	promote	Lantern consolidate/offline token rule	separates owned compute from cloud billing
Keep Archive/Wayback media metadata-first	promote/hold	archive commons policy	rights and storage review required
Hold v1.0.0 release	hold	readiness gates	operator approval not yet given

STRENGTHS

- 1. Local-first architecture is now explicit.
- 2. RAG memory has a single flat canonical file.
- 3. Assets are copied intentionally and hashed.
- 4. Whitepaper exists in Markdown and PDF.
- 5. Cash sprint has an invoice and wallet ledger without fake balance.
- 6. Dual boot has clear readiness evidence and a safe physical boundary.
- 7. Skills now route repeatable work instead of relying on chat memory.
- 8. Governance is modeled as evidence-backed offchain review, not one-shot impulse.

RISKS

Risk	Severity	Current Control	Next Fix
Physical disk mutation during dual boot	high	hard boundary in docs/scripts	operator-only Disk Management step
Fake revenue or blurred wallet state	high	cleared cash remains \$0	record sent/paid events only when real
Repo sprawl returning	medium	master repo and RAG flat file	update dollhouse after every sprint
Orchestrator dirty state	resolved	committed and pushed f4eb6b5	keep supervisor health checks responsive
External media rights	medium	metadata-first commons lane plus rights gate	operator verifies rights before downloads
Whitepaper overclaiming	medium	v0.1 draft status and boundaries	revise after first cash result
V1 release too early	high	release gate hold	operator says promote only after gates pass

ADS GATE REVIEW

Gate	Status	Evidence	Action
Repo Cleanliness	pass	master...origin/master, no local changes before this review	keep small commits
RAG Integrity	pass	42 copied assets, SHA manifest rebuilt	rehash after any asset add
Whitepaper	pass	Markdown and PDF built	use as canonical public artifact
Wallet	pass	local wallet JSON and JSONL parse	record only factual events
Cash Sprint	candidate/send-ready	P0 orch task queued, send packet and invoice draft exist	send/record 5 warm messages
Dual Boot Prep	candidate	prep-ready, no failures	run elevated and shrink D: manually
Dual Boot Install	held	0 GB unallocated space	operator physical action
Commons Batch	candidate/held-downloads	metadata script and rights gate exist	downloads need operator rights/storage review
v1.0.0	held	operator approval missing	no release/tag yet

CURRENT DUAL BOOT ADS FINDING

The machine is not blocked by capability. It is blocked by prepared space.

```
readyForPrep: true
readyForInstall: false
primary blocker: 0.0 GB unallocated install space
D: free: 1636.9 GB of 1863.0 GB
```

ADS decision: promote preparation, hold installation. Preparation is now scripted by dual-boot/Start-DualBootPrep.ps1.

Next physical step:

```
Run elevated readiness -> backup/recovery keys -> shrink D: 100-250GB -> leave unallocated -> rerun readiness.
```

CURRENT WALLET ADS FINDING

The wallet architecture is correct for now because it prevents fantasy money.

```
clearedCashUsd: 0
pendingInvoiceUsd: 199
firstInvoice: INV-COMET-LEAP-RAG-001
```

ADS decision: promote local ledger and send packet, hold payment-provider integration until a real buyer needs a real payment path.

Send-ready packet:

```
data/cash-loop/OUTREACH-SEND-PACKET.md
```

Wallet event logger:

```
scripts/Add-WalletLedgerEvent.ps1
```

CURRENT RAG ADS FINDING

The RAG house is the correct memory spine because it makes every major artifact findable from one file.

Current anchor:

```
skills/lantern-rag-dollhouse/references/LANTERN-OS-RAG-DOLLHOUSE.flat.md
```

ADS decision: promote. Every future sprint should update the flat file and asset manifest before pushing.

IMMEDIATE NEXT ACTIONS

- 1. Print or share artifacts/LANTERN-OS-WHITEPAPER-v0.1.pdf.
- 2. Execute the P0 orchestrator cash-loop task.
- 3. Record 5 outreach attempts or 5 explicit contact blockers.
- 4. If one buyer says yes, send the \$199 RAG cleanup invoice draft.
- 5. Run elevated dual-boot readiness.
- 6. Shrink D: manually by 100-250GB only after backup/key checks.
- 7. Rerun readiness and save the result.
- 8. Update this ADS review after the first cash outcome or dual-boot prep change.

BLOCKER FIX PASS

The operator-requested blocker fix pass created:

- dual-boot/Start-DualBootPrep.ps1;
- data/cash-loop/OUTREACH-SEND-PACKET.md;
- scripts/Add-WalletLedgerEvent.ps1;
- data/archive-commons/RIGHTS-REVIEW-GATE.md;
- manifests/BLOCKER-FIX-2026-05-26.md;
- orchestrator fix commit f4eb6b5.

ARCHITECTURE VERDICT

Lantern OS is architecturally converged enough to operate.

It is not release-final, but it is now action-ready:

```
ship whitepaper -> run cash sprint -> record wallet events -> prepare dual boot -> update RAG house -> repeat
```